The House Is You: Why Strong Players Underwager on Jeopardy!

Version 2.1 — 2026-10-13. A thesis: an argument with its evidence, its counter-evidence and the tests it still needs. Model figures come from this project's engine at version 2.1.0; the white paper documents that engine.

Abstract

The claim: careful, intelligent people learn early that gambling is a losing game, because the house always wins. On Jeopardy! they are no longer the customer. When they know a category, a Daily Double is a bet with the odds in their favor, so they are the house. Many strong players still bet as if they were the customer. They make small, "safe" wagers when the game rewards large ones.

The evidence partly supports this and partly does not. Published data shows that human Daily Double wagers fall well below the win-maximizing amount. This project's model shows the cost is large for a trailing player and small for a leader. But the premise that intelligent people avoid risk in general is contradicted by the best-known study of the question. A narrower claim survives: a heuristic that is correct in a casino gets applied to a game where it is wrong. Tournament formats such as Jeopardy! Masters change the incentives, and some caution there is rational.

1. The claim, stated so it can be wrong

Three parts:

1. **Heuristic.** People who avoid gambling hold a correct rule for negative-expectation games: "the more you bet, the more you lose."
2. **Transfer.** On Jeopardy! they apply the same rule to bets with positive expectation, and wager less than they should.
3. **Cost.** The underwager costs real win probability, most of all when the player is behind.

Each part can fail on its own. Section 3 tests part 1, section 4 part 2, and section 5 part 3.

2. Why "the house always wins" stops being true

A casino bet has negative expectation for the player. In American roulette, a single-number bet pays 35 to 1 on a 1-in-38 chance. The expected return per dollar is (35 - 37)/38 = -2/38, a 5.26% house edge. Every dollar wagered loses 5.26 cents on average, so the rational amount to bet is zero.

A Daily Double reverses the sign. With wager w and probability p of a correct response, the expected change in score is

```
E[gain] = p·w − (1 − p)·w = (2p − 1)·w
```

When p > 0.5, every extra dollar wagered *adds* (2p - 1) dollars in expectation. At 70% confidence that is +40 cents per dollar: a 40% edge, in the player's favor, nearly eight times the roulette house edge. The player holds the edge the casino normally holds. On this measure alone, "bet less" is backwards.

Two cautions keep this honest:

- Jeopardy! does not reward expected dollars. In regular play it rewards *winning the game* (the winner keeps their total and returns). The right target is win probability, which is what this project's engine maximizes. Win probability and expected dollars usually point the same way when you are behind, and can point opposite ways when you lead.
- p is the player's own estimate. An underconfident player bets less for a reason that is not risk aversion at all.

3. Do intelligent people avoid risk? The counter-evidence

The premise in part 1 is that intelligent people are risk-averse. The best-known direct test finds the opposite. Dohmen, Falk, Huffman and Sunde (2010) ran incentivized risk-choice experiments with about 1,000 adults in a representative German sample. They found that **lower** cognitive ability goes with **more** risk aversion, after controls for education, income and credit constraints. The result is correlational.

So "smart people avoid risk" is not supported as a general trait. The claim must be narrower to survive:

- **Domain-specific rule, not a trait.** People can be willing to take positive-expectation risks in a lab and still hold a firm rule about *gambling*, a domain they have learned is a losing one. Jeopardy! wagering looks and feels like gambling: a dollar amount, a stake, a reveal.
- **Loss aversion.** Losses weigh more than equal gains (Kahneman and Tversky, 1979). On television, a loss is also public and permanent.
- **Regret.** A big wager that misses is remembered as a mistake. A small wager that hits is not remembered as money left on the table.

None of these three is measured in this document. They are candidate mechanisms, not findings.

4. Evidence that players underwager

Tesauro, Gondek, Lenchner, Fan and Prager (2013) studied about 3,000 real games from the mid-1990s to 2012. In last-Daily-Double situations in the second round, they report:

Situation	Average human wager	Wager their model recommends
Player leading	$2,590	$5,380
Player trailing	$3,120	$6,970

Humans bet about half of the recommended amount, in both positions. This is the most direct published evidence for part 2 of the claim. It describes play before the Daily Double hunting era. Since then, players have bet Daily Doubles much harder (see ADR-0004), so the gap has likely narrowed. This document does not measure by how much: the modern statistics come from a source whose terms forbid republishing them.

Jeopardy! Masters. The claim that elite players underwager on the Masters specials is an observation, not a measurement here. Testing it needs Masters wager data, which this project does not publish (see section 7).

5. Measured: what underwagering costs

Method: this project's Daily Double engine (`engine/dd_wager.py` , version 2.1.0) simulates the rest of the game against simulated modern-era rivals, then Final Jeopardy, and scores a tie for first as a share of a win. The state below is a Double Jeopardy! Daily Double with about $8,000 left on the board and no Daily Doubles after it. Each row: 2,000 simulated games per wager, wagers on a $200 grid, two seeds (7 and 8); the figure is the range across the two seeds. "Human average" uses Tesauro's published averages from section 4 as the reference wager.

Trailing player (scores $6,000 vs $12,000 vs $8,000, you have $6,000):

Confidence	Bet $2,000	Human average ($3,120)	Bet everything ($6,000)
70%	15%	16-17%	**26%**
60%	14%	14-15%	**22%**

Betting everything is the engine's recommendation in every run, and the only wager on its plateau. The typical human bet gives up about 9 to 10 points of win probability at 70% confidence: nearly 40% of the player's chances.

Leading player (scores $12,000 vs $8,000 vs $6,000, you lead):

Confidence	Bet $2,000	Human average ($2,590)	Recommended-size ($5,380)	Best estimate
70%	67%	69%	70-72%	72-73% (at $7,400-$7,800)
60%	64%	65%	63-65%	65-66% (flat; $0 to ~$10,000 not distinguishable)

For the leader the cost is a few points at 70%, and at 60% a small bet is as good as anything. The engine's statistical plateau is wide here: many wagers cannot be told apart at this sample size.

Final Jeopardy!, for contrast (Tier 2 engine, exact; you lead $12,000 vs $8,000 vs $6,000):

Confidence	Bet $0	Cover bet ($4,001)
90%	58.0%	**95.9%**
70%	58.0%	**85.8%**

A leader who "plays it safe" by betting nothing gives up 28 to 38 points of win probability. The cover bet is the safe bet.

Commands: see section 9.

6. When betting less is right

The thesis does not say "always bet more." The same engine says caution is correct in several places:

- **Leading with modest confidence.** At 60% confidence the leader's curve is flat (section 5): extra wager buys little.
- **Second place in Final Jeopardy!.** With $8,000 behind a $12,000 leader (third place $6,000) at 70% confidence, the engine recommends $0 (36.9% win share). Betting small and hoping the leader misses is a real strategy.
- **Tournament incentives.** Jeopardy! Masters awards match points per game: 3 for first, 1 for second, 0 for third, with dollar totals used only in the two-game final (format as reported for recent seasons; check the official rules for a given year). Second place is worth something, so a bet that risks dropping from second to third has a real cost that regular play does not have. Some Masters caution is rational play under those rules, not timidity. This weakens Masters as evidence for the thesis: an underwager there must be judged against a match-points objective, which this project's engine does not model.

7. Data used, and what was not used

- **Published:** Tesauro et al. (2013), regular episodes, mid-1990s to 2012.
- **Model calibration:** the engine's simulated rivals are calibrated privately to J! Archive season statistics for Seasons 38-41 (ADR-0004). Those season pages include the in-season tournaments (for example, the Tournament of Champions) as well as regular play, so a minority of the calibrating games had tournament incentives. Jeopardy! Masters is archived separately and was **not** used. No J! Archive figure appears here; its terms forbid republication.
- **Not used:** any per-player, per-game wager record, and any data on contestants' education, intelligence or attitude to gambling.

8. What this does not prove

- It does not show that *intelligent* players underwager more than others. There is no measure of intelligence here, and the best evidence (section 3) says ability goes with less risk aversion, not more.

- It does not show *why* players underwager. Heuristic transfer, loss aversion, regret and underconfidence all predict small bets; nothing here separates them.
- The cost figures in section 5 are model outputs, not observed outcomes. They depend on the engine's assumptions (simplified regular-clue play, simulated rivals, the tiebreaker scoring). They have not been checked against real games.
- The human-average wagers are from before about 2012. Modern players bet harder; the size of today's gap is unmeasured here.
- The Masters observation is not tested at all.

9. How to test it properly

1. **Gap, modern era.** For each real Daily Double, compare the actual wager with the engine's recommendation for that state, split by position (leading or trailing). Needs per-game data with a license that permits publication.
2. **Who underwagers.** Link wager gaps to a proxy for ability (for example, regular-clue accuracy in the same game) to test the "smart players" part directly, controlling for score position.
3. **Mechanism.** Underconfidence predicts a gap that shrinks in categories the player answers well; loss aversion predicts a gap that grows with the stake. The two can be separated by category and score.
4. **Masters.** Rebuild the objective as expected match points, then judge Masters wagers against that objective, not against win probability.

Reproduce section 5 (from `engine/`, version 2.1.0):

```
import dd_wager, fj_bestresponse as br
from simulate import RolloutConfig
cfg = RolloutConfig(remaining_clue_values=[1100] * 7, remaining_dd_count=0)
r = dd_wager.analyze([6000, 12000, 8000], 0, 2, 0.7, config=cfg,
                     wager_step=200, trials_per_wager=2000, seed=7)
print(r.recommended_wager, dict(r.wager_win_probabilities)[6000])
print(br.analyze([12000, 8000, 6000], 0, 0.9).win_probability)
```

References

- Dohmen, T., Falk, A., Huffman, D., & Sunde, U. (2010). Are risk aversion and impatience related to cognitive ability? *American Economic Review*, 100(3), 1238-1260.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263-291.
- Tesauro, G., Gondek, D., Lenchner, J., Fan, J., & Prager, J. M. (2013). Analysis of Watson's strategies for playing Jeopardy!. *Journal of Artificial Intelligence Research*, 47, 205-251. https://arxiv.org/pdf/1402.0571
- Jeopardy! Masters format (match points 3/1/0): TV Insider, "Jeopardy! Masters 2025: matchups, schedule, players, and more." https://tvinsider.com/1184784/jeopardy-masters-2025-prize-cast-premiere-date-more

- This project: `docs/white-paper.md` (the engine and its figures), `docs/adr/0004-modern-era-calibration.md` (the calibration).